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Densities and derived thermodynamic properties for the (*n*-heptane + *n*-octane), (*n*-heptane + ethanol) and (*n*-octane + ethanol) systems at high pressures

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ABSTRACT

This work reports densities for the pure *n*-heptane, *n*-octane and ethanol at temperatures from (288.15 to 413.15) K and at pressures up to 60 MPa. The same properties were determined for the binary mixtures *n*-heptane + *n*-octane, *n*-heptane + ethanol and *n*-octane + ethanol over the temperature range (293.15–373.15) K and at pressures ranging from (0.1 to 40) MPa. Highly precise densimeter DMA HP was used for density measurements. Experimental values of densities were fitted by the modified Tammann–Tait equation. Additionally, derived volumetric properties, e.g. excess molar volumes, isothermal compressibilities and isobaric thermal expansion coefficients were calculated. The effect of pressure and temperature on the measured and derived properties was discussed.

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1. Introduction

In order to reduce the greenhouse gases, especially CO_2 emissions, the incorporation of non-fossil compounds in automotive fuels is of high interest nowadays. Worldwide, mixtures of ethanol and conventional gasoline are used due to many positive aspects of ethanol application: it has a good "well to tank" CO_2 emission balance since it is extracted from the biomass, high octane number (108), high heat of vaporization, broader flammability limits and higher flame speeds [1,2]. On the other hand, linear alkanes are typical gasoline constituents so thermodynamic and mechanical properties of liquid alkane + alcohol or alkane + alkane mixtures are of great interest, especially at elevated pressure and temperature conditions. From p, ρ, T, x data of liquid mixtures valuable derived properties can be calculated, e.g. isothermal compressibilities and isobaric thermal expansion coefficients. All these properties are essential for the design, operation, control and optimization of industrial processes.

This work presents a part of our ongoing research related to the determination of densities and derived thermodynamic properties at high pressures for pure compounds, binary and multicomponent mixtures of gasoline constituents and of biodiesel [3–5]. Here, results of the density measurements of pure *n*-heptane, *n*-octane and ethanol at temperatures from (288.15 to 413.15) K and at pressures up to 60 MPa are reported. Also, densities of binary mixtures (*n*-heptane + *n*-octane), (*n*-heptane + ethanol) and (*n*-octane + ethanol) for the mole compositions about 0.25, 0.50 and 0.75 were determined over the temperature range (293.15–373.15) K at pressures ranging from (0.1 to 40) MPa. Modified Tammann–Tait equation was applied to correlate density data. From the measured densities derived thermodynamic properties were calculated and analyzed, excess molar volumes, V^E, isothermal compressibilities, κ_T, and isobaric thermal expansion coefficients, α_p.

Densities of the *n*-heptane + ethanol and *n*-heptane + *n*-octane mixtures have been investigated before [6–11] using different measuring techniques, while to the best of our knowledge density data of *n*-octane + ethanol at high pressures and temperatures have not been published before. Ozawa et al. [6] determined densities of *n*-heptane + ethanol at pressures up to 196.2 MPa over the temperature range (298.2–348.2) K, while Papaioannou et al. [7] performed measurements at 298.15 K and for pressures up to 33.81 MPa. Ulbig et al. [8] reported excess volumes of the *n*-heptane + ethanol system at 298.15 K and at pressures up to 60 MPa. Later on, Dzida and Marczak [9] obtained densities for the same system from speeds of sound measurements at pressures up to 90 MPa within the temperature limits (293 and 318) K. Watson et al. [10] determined densities of seven ethanol + *n*-heptane mixtures in the range (293.15–333.15) K and up to 65 MPa using vibrating-tube densimeter. Abdulagatov and Azizov [11] reported densities of three *n*-heptane + *n*-octane binary mixtures over the temperature range (293–557) K and at pressures up to 10 MPa.

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This work provides high pressure density data for the *n*-octane + ethanol system that has not been analyzed in the literature. Furthermore, densities and derived property data for the *n*-heptane + ethanol and *n*-heptane + *n*-octane mixtures reported here cover some temperature or pressure conditions that have not been studied in the literature.

2. Materials and methods

2.1. Materials

The chemicals in this work were used as received without further purification. *n*-Heptane was supplied from Sigma–Aldrich, while *n*-octane and ethanol were the products of Merck, all having the mass fraction purities ⩾0.99. Chemicals were degassed prior to measurements. Details of chemicals including their certified purities are listed in Table 1. Table 2 gives comparison of densities of the pure compounds, measured by using DMA 5000 densimeter [12], with literature values at atmospheric pressure.

2.2. Apparatus

The liquid densities of pure compounds and its binary mixtures at different pressures and temperatures were experimentally studied using an Anton Paar DMA HP density measuring cell coupled with an Anton Paar DMA 5000 vibrating tube densimeter. The measuring device was explained in details in our previous works [3,5].

The apparatus was used to measure the oscillating period for the pure compounds over the temperature range (288.15–413.15) K and at pressures from (0.1 to 60) MPa and for binary mixtures at temperatures from (293.15 to 373.15) K and at pressures up to 40 MPa. Each selected temperature was controlled with an integrated Peltier thermostat with the stability within ±0.05 K. A pressure generator, model 50-6-15 from High Pressure Equipment Co. (HiP), was used to set and control the pressure in the system; acetone was applied as a hydraulic fluid [3]. Pressure in the system was measured using a pressure transducer WIKA, S-10, Alexander WiegandGmbH&Co. The maximum value of the sums of deviation and expanded measurement uncertainty is ±0.1815% in relation to the full range.

The period of the tube vibration τ, was determined with a digital counter whilst the vibrating period was displayed to seven significant digits. The collected data (period of oscillation of U-tube, cell temperature and pressure) were transferred from DMA HP to DMA 5000 by connecting S-BUS interfaces. The APSoftPrint software program (a Microsoft Excel Add-In) was used to read out and transfer the measured values from DMA 5000 to a PC.

The procedure of Comuñas et al. [16] has been employed to calibrate the Anton Paar DMA HP densimeter and described in details elsewhere [3].The influence of sample' s viscosity on density was estimated by using the information retrieved from a personal correspondence with the supplier Anton Paar and it was less than 0.02 kg · m^{-3}. Having in mind the uncertainties of pressure, temperature, period of oscillation measurements and the damping effects on the vibrating tube, the overall expanded uncertainty with 0.95 level of confidence (k = 2) in the reported densities is estimated to be 1.7 kg · m^{-3} in the temperature range (288.15–363.15) K and 2.7 kg · m^{-3} at temperatures (373.15–413.15) K.

Mixtures were prepared gravimetrically using a Mettler AG 204 balance with a precision $1 \cdot 10^{-7}$ kg. The uncertainty of the mole fraction calculation was less than $\pm 1 \cdot 10^{-4}$. The excess molar volume was calculated with the average uncertainty $\pm 4 \cdot 10^{-6}$ m^3 · kmol^{-1}.

Table 1
Chemicals used in this work.

Chemical name	CAS	Supplier	Purity mass fraction
n-Heptane	142-82-5	Sigma–Aldrich	⩾0.99
n-Octane	111-65-9	Merck	⩾0.99
Ethanol	64-17-5	Merck	⩾0.999

Table 2
Comparison of the measured and literature densities of the pure compounds at atmospheric pressure.

Component	T^a/K	ρ^b/kg · m^{-3} (p^c = 0.101 MPa)	
		Exp.	Exp.
n-Heptane	288.15	688.33	687.92 [13]
	293.15	684.11	683.9 [10]; 683.67 [13]
	298.15	679.87	679.48 [13]
	303.15	675.60	675.1 [10]; 675.15 [13]
	308.15	671.31	670.86 [13]
	313.15	667.00	666.4 [10]; 666.5 [13]
n-Octane	293.15	702.58	702.56 [13]; 702.55 [14]
	303.15	694.48	694.48 [13]; 694.44 [14]
	313.15	686.32	686.29 [13]; 686.22 [14]
	323.15	678.09	678.00 [13]; 677.90 [14]
Ethanol	303.15	780.83	781.1 [10], 780.9 [15]
	313.15	772.10	772.3 [10], 772.1 [15]
	323.15	763.22	763.1 [10], 763.3 [15]

[a] $u(T)$ = 0.01 K (standard uncertainty).
[b] $U(\rho)$ = 0.08 kg · m^{-3}(expanded uncertainty with 0.95 level of confidence).
[c] $u(p)$ = 5% (standard uncertainty).

3. Results and discussion

3.1. Measured densities

Densities of the pure *n*-heptane, *n*-octane and ethanol were determined for temperatures between (288.15 and 413.15) K and at pressures ranged from (0.1 to 60) MPa. The experimental results are presented in Table 3 and some selected results are shown graphically in Figs. S1 and S2. Densities of the pure compounds measured here are compared with literature values. An absolute average percentage deviation ($AAD(\%) = 100/N\sum_{i=1}^{N}|(\rho_i^{lit}/\rho_i^{exp} - 1)|$) and maximum percentage deviation ($MxD(\%) = \max(100|(\rho_i^{lit}/\rho_i^{exp} - 1)|)$) are used to estimate the agreement between our and literature data. Fig. 1 shows comparison between densities of the pure compounds measured here and those previously reported. In some cases, it was necessary to interpolate the literature values in order to obtain the data that correspond to certain pressures and temperatures considered here.

The absolute average percentage deviation between the present densities of pure *n*-heptane and data of Sanmamed et al. [13] and Lampreia and Nieto de Castro [14], both at temperatures from (288.15 to 323.15) K, and Watson et al. [10] within temperature range (293.15–333.15), are 0.039%, 0.037%, and 0.027%, respectively, all at pressures up to 60 MPa (Fig. 1a). The data of Dzida and Marczak [9] deviate from our experimental results with AAD = 0.080% at temperatures (293.15–318.15) K and densities of Toscani et al. [17] deviate with AAD = 0.137% at temperatures (298.15–373.15) K, also at pressures up to 60 MPa (Fig. 1a). Absolute average deviation of our and densities calculated using correlation proposed by Cibulka and Hnědkovský [18], over the temperature range (288.15–363.15) and at pressures up to 60 MPa is 0.064% (Fig. 1b). Literature densities, measured at temperatures other than those reported here, were compared to densities from this work calculated using the modified Tammann–Tait equation (see Section 3.2). The agreement of these densities with

Table 3
Experimental densities of pure compounds at various pressures, p, and temperatures, T.

p[c]/MPa	ρ[a]/kg · m^{-3}							
	T[b]/K							
	288.15	293.15	298.15	303.15	308.15	313.15	318.15	323.15
n-Heptane								
0.1	688.4	684.2	679.9	675.7	671.4	667.1	662.7	658.3
1	689.2	685.0	680.7	676.5	672.2	668.0	663.6	659.3
5	692.5	688.4	684.3	680.2	676.0	672.0	667.7	663.5
10	696.5	692.6	688.6	684.6	680.6	676.7	672.6	668.6
15	700.4	696.5	692.7	688.9	685.0	681.2	677.3	673.4
20	704.1	700.4	696.6	693.0	689.2	685.5	681.8	678.0
25	707.7	704.0	700.4	696.8	693.2	689.6	686.0	682.4
30	711.1	707.5	704.0	700.5	697.0	693.5	690.0	686.5
35	714.3	710.9	707.4	704.0	700.6	697.2	693.8	690.4
40	717.4	714.0	710.7	707.4	704.0	700.7	697.4	694.1
45	720.3	717.0	713.8	710.5	707.2	704.0	700.7	697.5
50	723.1	719.9	716.7	713.5	710.3	707.1	703.9	700.7
55	725.8	722.5	719.4	716.2	713.1	709.9	706.8	703.7
60	728.2	725.1	721.9	718.8	715.7	712.6	709.5	706.4
	328.15	333.15	343.15	353.15	363.15	373.15	393.15	413.15
0.1	653.9	649.5	640.5	631.3	621.9			
1	654.9	650.6	641.6	632.5	623.2	614.2	594.6	574.1
5	659.3	655.1	646.5	637.8	628.9	620.2	601.7	582.3
10	664.5	660.5	652.3	644.0	635.6	627.4	610.0	592.0
15	669.5	665.7	657.8	649.9	641.9	634.1	617.8	600.9
20	674.3	670.6	663.1	655.5	647.9	640.4	625.0	609.2
25	678.8	675.2	668.0	660.7	653.4	646.3	631.7	616.9
30	683.0	679.6	672.6	665.6	658.6	651.8	637.9	623.9
35	687.0	683.7	676.9	670.2	663.5	656.9	643.6	630.2
40	690.8	687.5	681.0	674.4	667.9	661.5	648.7	635.9
45	694.3	691.1	684.7	678.3	672.0	665.8	653.3	640.9
50	697.6	694.4	688.1	681.9	675.7	669.6	657.4	645.2
55	700.6	697.4	691.3	685.1	679.0	673.1	660.9	648.9
60	703.3	700.2	694.1	688.0	681.9	676.1	663.9	651.9
n-Octane								
	288.15	293.15	298.15	303.15	308.15	313.15	318.15	323.15
0.1	706.5	702.5	698.5	694.4	690.4	686.3	682.2	678.0
1	707.2	703.2	699.2	695.2	691.2	687.1	683.0	678.9
5	710.3	706.4	702.5	698.6	694.7	690.8	686.8	682.8
10	714.1	710.3	706.5	702.7	698.9	695.1	691.3	687.4
15	717.7	714.0	710.3	706.7	703.0	699.3	695.6	691.9
20	721.2	717.6	714.0	710.5	706.9	703.3	699.7	696.1
25	724.5	721.0	717.6	714.1	710.6	707.2	703.6	700.2
30	727.7	724.3	720.9	717.5	714.2	710.8	707.4	704.0
35	730.8	727.5	724.1	720.9	717.5	714.3	711.0	707.7
40	733.7	730.5	727.2	724.0	720.8	717.6	714.3	711.1
45	736.5	733.3	730.1	727.0	723.8	720.7	717.5	714.4
50	739.2	736.0	732.9	729.8	726.7	723.6	720.5	717.4
55	741.7	738.6	735.5	732.5	729.4	726.3	723.3	720.3
60	744.1	741.0	738.0	735.0	731.9	728.9	725.9	722.9
	328.15	333.15	343.15	353.15	363.15	373.15	393.15	413.15
0.1	673.8	669.7	661.2	652.6	643.8	634.7	616.3	
1	674.7	670.6	662.2	653.7	645.0	636.3	618.1	599.3
5	678.7	674.8	666.6	658.4	650.1	641.8	624.3	606.6
10	683.5	679.7	671.9	664.1	656.1	648.2	631.8	615.1
15	688.1	684.5	677.0	669.4	661.9	654.4	638.8	623.1
20	692.5	689.0	681.8	674.5	667.3	660.1	645.3	630.6
25	696.7	693.2	686.3	679.3	672.4	665.5	651.5	637.4
30	700.7	697.3	690.6	683.9	677.2	670.6	657.1	643.8
35	704.4	701.1	694.6	688.1	681.6	675.3	662.3	649.6
40	707.9	704.7	698.4	692.1	685.8	679.6	667.1	654.8
45	711.3	708.1	701.9	695.7	689.6	683.6	671.4	659.5
50	714.4	711.3	705.2	699.1	693.1	687.3	675.3	663.6
55	717.3	714.2	708.2	702.3	696.3	690.5	678.8	667.2
60	719.9	716.9	711.0	705.1	699.2	693.5	681.8	670.2
Ethanol								
	288.15	293.15	298.15	303.15	308.15	313.15	318.15	323.15
0.1	793.7	789.4	785.2	780.9	776.5	772.1	767.7	763.2
1	794.4	790.2	785.9	781.6	777.3	773.0	768.5	764.1
5	797.6	793.5	789.3	785.1	780.9	776.6	772.3	768.0
10	801.5	797.4	793.4	789.3	785.2	781.0	776.8	772.6

(*continued on next page*)

Table 3 (*continued*)

p[c]/MPa	ρ[a]/kg · m^{-3}							
	T[b]/K							
	288.15	293.15	298.15	303.15	308.15	313.15	318.15	323.15
15	805.3	801.3	797.3	793.3	789.3	785.3	781.2	777.1
20	808.9	805.0	801.1	797.2	793.3	789.4	785.4	781.4
25	812.4	808.6	804.8	801.0	797.1	793.3	789.4	785.5
30	815.7	812.0	808.3	804.6	800.8	797.0	793.2	789.4
35	819.0	815.3	811.7	808.0	804.3	800.6	796.9	793.2
40	822.1	818.5	814.9	811.3	807.7	804.1	800.4	796.8
45	825.1	821.5	818.0	814.5	810.9	807.4	803.8	800.2
50	828.0	824.5	821.0	817.5	814.0	810.5	806.9	803.4
55	830.7	827.2	823.8	820.3	816.9	813.4	809.9	806.5
60	833.3	829.9	826.5	823.1	819.6	816.2	812.8	809.3
	328.15	333.15	343.15	353.15	363.15	373.15	393.15	413.15
0.1	758.7	754.1	744.6					
1	759.6	755.0	745.6	736.0	725.9	715.3	692.2	666.3
5	763.6	759.1	750.0	740.6	730.8	720.6	698.4	673.6
10	768.3	764.0	755.2	746.2	736.7	727.0	705.8	682.3
15	772.9	768.8	760.2	751.5	742.4	733.0	712.8	690.5
20	777.3	773.3	765.0	756.5	747.7	738.7	719.4	698.1
25	781.6	777.6	769.6	761.3	752.8	744.2	725.6	705.3
30	785.6	781.7	773.9	765.9	757.7	749.3	731.4	712.0
35	789.5	785.7	778.0	770.2	762.2	754.1	736.8	718.2
40	793.1	789.4	781.9	774.3	766.5	758.6	741.8	723.8
45	796.6	792.9	785.6	778.1	770.5	762.8	746.4	729.0
50	799.9	796.3	789.1	781.7	774.3	766.7	750.7	733.7
55	803.0	799.4	792.3	785.1	777.7	770.2	754.5	737.9
60	805.9	802.4	795.3	788.2	780.9	773.5	757.9	741.5

[a] $U(\rho)$ = 1.7 kg · m^{-3} (288.15 K ⩽ T ⩽ 363.15 K) and 2.7 kg · m^{-3} (373.15 K ⩽ T ⩽ 413.15 K) (expanded uncertainty with 0.95 level of confidence).
[b] $U(T)$ = 0.1 K (expanded uncertainty with 0.95 level of confidence).
[c] $U(p)$ = 0.05 MPa (expanded uncertainty with 0.95 level of confidence).

data reported by Pečar and Doleček [19], at temperatures (298.15–348.15) K, and with Morávková and Linek [20], over the temperature range (298.15–328.15) K, both at pressures up to 40 MPa, is within 0.063% and 0.039%, respectively. Absolute average deviation between data of Quevedo-Nolasco et al. [21] and our calculated densities, at temperatures (313.15–353.15) K and up to 25 MPa, is 0.049% (Fig. 1b). Good agreements of our data at atmospheric pressure and those reported by Matos et al. [22] in the temperature range (293.15–313.15) K, with *AAD* = 0.064%, and Abdulagatov and Azizov [11] at temperatures (303.15–353.15) K with *AAD* = 0.026%, were achieved. Deviation between our and data of Papaioannou et al. [7] at 298.15 K and pressures up to 33.8 MPa were 0.056% (Fig. 1b).

Average absolute deviation for *n*-octane comparing our and the results of both Sanmamed et al. [13] and Lampreia and Nieto de Castro [14] are 0.086%, over the temperature range (288.15–323.15) K and at pressures up to 60 MPa. The agreement between our and data of Morávková et al. [23], at temperatures (298.15–328.15) K and at pressures up to 40 MPa, is a bit better, *AAD* = 0.043% (Fig. 1c). The agreement of the densities presented here and those calculated using method of Cibulka and Hnědkovský [18] is 0.064%, over the temperature range (288.15–393.15) K and at pressures of interest in this work. Lugo et al. [24] reported densities of pure *n*-octane at temperatures (293.15–353.15) K and pressures up to 25 MPa which agree with our densities within 0.043%, and densities of Caudwell et al. [25] match ours, calculated using modified Tammann–Tait equation (see Section 3.2), with *AAD* = 0.109% at temperatures (298.15–398.15) K up to 60 MPa (Fig. 1c). Matos et al. [22], Abdulagatov and Azizov[11] and Estrada-Baltazar et al. [26] reported densities of *n*-octane at atmospheric pressure and at the temperature ranges (293.15–413.15) K, (303.15–353.15) K and (293.15–323.15) K, that fit to our data within 0.012%, 0.026% and 0.006%, respectively (Fig. 1c).

For the pure ethanol our results agree with literature within 0.058%, 0.082% and 0.065% compared to Watson et al. [10], over the temperature range (293.15–333.15) K, to Dzida and Marczak [9], at temperatures (293.15–318.15) K, and to correlation proposed by Cano-Gómez et al. [27], for temperatures between 293.15 K and 343.15 K, all at pressures up to 60 MPa, respectively. Zéberg-Mikkelsen et al. [28,29] reported ethanol density data at temperatures (293.15–353.15) K and pressures up to 45 MPa which deviate from our densities within 0.027%, and comparison of our densities with density values of Lasarte et al. [30] resulted in an *AAD* = 0.040%, at temperatures (288.15–313.15) K and pressures up to 20 MPa (Fig. 1d). Ethanol densities of Ozawa et al. [6], at temperatures (298.2–348.2) K and pressures up to 58.9 MPa, of Ott et al. [31] at temperatures (298.15–348.15) K up to 15 MPa, and data of Takiguchi et al. [32], at temperatures (310–380) K up to 60 MPa, agree with our values calculated using modified Tammann–Tait equation (see Section 3.2) with somewhat higher *AAD* of 0.118%, 0.215% and 0.131%, respectively (Fig. 1d). Additionally, our experimental data show deviations of 0.188% and 0.014% comparing to densities reported by Estrada-Baltazar et al. [26] (at atmospheric pressure and temperatures (293.15–413.15) K) and by Papaioannou et al. [7] at 298.15 K and pressures (0.1–33.8) MPa (Fig. 1d).

Fig. S1 shows measured densities of the pure compounds as a function of temperature at different pressures. Fig. S1 demonstrates that densities are decreasing in the following order: ethanol > *n*-octane > *n*-heptane. As temperature increases a pure compound density decreases almost linearly at investigated temperature and pressure ranges. Contrarily, density increases with increasing pressure. The variation of density versus pressure at lower temperatures is practically linear, while it slightly deviates from straight line at temperatures higher than 353.15 K (Fig. S2).

Densities of the binary mixtures *n*-heptane + *n*-octane, *n*-heptane + ethanol and *n*-octane + ethanol for the mole compositions about

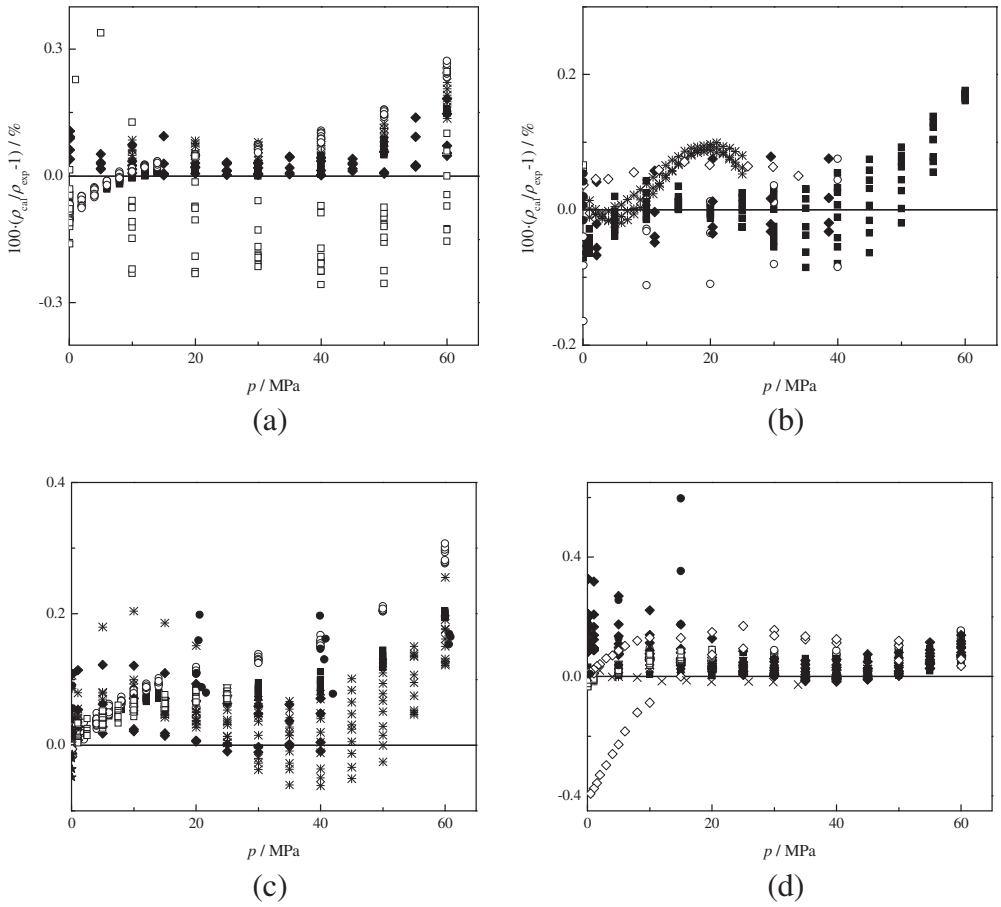


Fig. 1. Comparison between measured densities and those previously reported against pressure for (a) *n*-heptane: (■) Sanmamed et al. [13] at (288.15–323.15) K, (○) Lampreia and Nieto de Castro [14] at (288.15–323.15) K, (◆) Watson et al. [10] at (293.15–333.15) K and (⋇) Dzida et al. [9] at (293.15–318.15) K and (□) Toscani et al. [17] at (298.15–373.15) K; (b) *n*-heptane: (■) Cibulka and Hnědkovský [18] at (288.15–363.15) K, (○) Pečar et al. [19] at (298.15–348.15) K, (◆) Morávková et al. [20] at (298.15–328.15) K, (⋇) Quevedo-Nolasco et al. [21] at (313.15–353.15) K, (□) Matos et al. [22] at (293.15–313.15) K, (●) Abdulagatov and Azizov [11] at (303.15–353.15) K and (◇) Papaioannou et al. [7] at 298.15 K; (c) *n*-octane: (■) Sanmamed et al. [13] at (288.15–323.15) K, (○) Lampreia and Nieto de Castro [14] at (288.15–323.15) K, (◆) Morávková et al. [22] at (298.15–328.15) K, (⋇) Cibulka and Hnědkovský [18] at (288.15–393.15) K, (□) Lugo et al. [23] at (293.15–353.15) K, (●) Caudwell et al. [24] at (298.15–398.15) K, (◇) Matos et al. [22] at (293.15–313.15) K and (★) Abdulagatov and Azizov [11] at (303.15–353.15) K and (×) Estrada-Baltazar et al. [26] at (293.15–323.15) K; and (d) ethanol: (■) Watson et al. [10] at (293.15–333.15) K, (○) Dzida et al. [9] at (293.15–318.15) K, (◆) Cano-Gómez [25] at (293.15–343.15) K, (⋇) Zéberg-Mikkelsen et al. [26,27] at (293.15–353.15) K, (□) Lasarte et al. [28] at (288.15–313.15) K, (●) Ott et al. [29] at (298.15–348.15) K, (◇) Takiguchi et al. [30] at (310–380) K, (★ Estrada-Baltazar et al. [26] at (293.15–323.15) K and (×) Papaioannou et al. [7] at 298.15 K.

0.25, 0.50 and 0.75 (see Tables 4–6) were measured over the temperature range (293.15–373.15) K and at pressures up to 40 MPa. Densities for the binary systems were compared with interpolated literature values where it was possible.

For *n*-heptane + *n*-octane absolute average percentage deviation between densities of this work and those interpolated from data of Abdulagatov and Azizov [11] was 0.057% at atmospheric pressure and at temperatures (303.15–333.15) K. Matos et al. [22] reported densities of *n*-heptane + *n*-octane mixtures at various compositions and *AAD* between interpolated and our data was 0.019% at temperatures (293.15–313.15) K and atmospheric pressure. The same procedure was applied to the data of Al Gherwi et al. [33], at 313.15 K and atmospheric pressure, which deviate from our densities within 0.020% (Fig. 2a).

For the *n*-heptane + ethanol mixtures *AAD* between our and the densities of Watson et al. [10] at temperatures (293.15–333.15) K at pressures up to 40 MPa was 0.025%. *AAD* of our experimental values and interpolated densities of Dzida and Marczak [9], at temperatures (293.15–313.15) K and pressures up to 40 MPa, was found to be 0.023%. Comparison with the densities of Papaioannou et al. [7] at 298.15 K and pressures up to 33.8 MPa and with Orge et al. [34] at atmospheric pressure and temperatures 303.15 K and 313.15 K resulted in *AAD* = 0.060% in 0.054%, respectively (Fig. 2b).

n-Octane + ethanol densities of Estrada-Baltazar et al. [26] were also interpolated and compared to our measured ones at atmospheric pressure and temperatures (293.15–323.15) K and obtained *AAD* was 0.080%. The comparison with interpolated values of Orge et al. [34] at atmospheric pressure and temperatures 303.15 K and 313.15 K showed *AAD* of 0.053% (Fig. 2c).

Fig. 3 depicts temperature and pressure influence on the densities of the investigated binary mixtures. Densities decrease as temperature increases. For the system *n*-heptane + *n*-octane experimental density vs. *x* data follow linear trend. Density vs. *x* function monotonically increases with increasing ethanol concentration along nonlinear dependence for the systems containing alcohol.

3.2. Density modeling

Experimental values of densities were fitted by modified Tammann–Tait equation. This correlation involves fitting parameters a_i, b_i and c_i and relates density ρ with temperature T and pressure p in a polynomial form according to the Eqs. (1)–(4)

Table 4
Experimental densities, ρ, for the binary mixture *n*-heptane (1) + *n*-octane (2) for different mole fractions, *x*, at pressure, *p*, and temperature, *T*.

p^c/MPa	ρ^a/kg · m^{-3}						
	n-Heptane (1) + *n*-octane (2)						
	T^b/K						
	293.15	303.15	313.15	323.15	333.15	353.15	373.15
x_1[d] = 0.2500							
0.1	698.22	690.1	681.8	673.4	664.9	647.5	
0.5	698.6	690.4	682.2	673.8	665.4	648.1	630.3
5	702.4	694.5	686.5	678.5	670.4	653.9	637.1
10	706.4	698.8	691.1	683.4	675.6	659.9	644.1
15	710.2	702.8	695.4	687.9	680.4	665.4	650.3
20	713.8	706.6	699.4	692.1	684.9	670.5	656.0
25	717.2	710.2	703.2	696.1	689.2	675.2	661.2
30	720.4	713.6	706.8	699.9	693.2	679.6	666.1
35	723.6	716.9	710.2	703.6	696.9	683.7	670.7
40	726.6	720.1	713.5	707.0	700.6	687.7	675.1
x_1 = 0.5002							
0.1	693.9	685.6	677.3	668.7	660.2	642.5	
0.5	694.3	686.0	677.7	669.2	660.6	643.1	625.0
5	698.1	690.2	682.1	674.0	665.8	649.1	632.0
10	702.2	694.5	686.8	678.9	671.1	655.2	639.2
15	706.0	698.6	691.1	683.6	676.0	660.8	645.6
20	709.7	702.5	695.2	687.9	680.6	666.0	651.4
25	713.2	706.1	699.1	692.0	684.9	670.8	656.7
30	716.5	709.6	702.7	695.8	689.0	675.3	661.7
35	719.6	712.9	706.2	699.5	692.8	679.5	666.3
40	722.7	716.1	709.6	703.0	696.5	683.6	670.8
x_1 = 0.7502							
0.1	689.0	680.7	672.2	663.5	654.8	636.9	
0.5	689.4	681.0	672.6	664.0	655.3	637.4	619.0
5	693.4	685.3	677.2	668.9	660.6	643.6	626.3
10	697.5	689.7	681.9	674.0	666.0	649.9	633.6
15	701.4	693.9	686.3	678.7	671.1	655.7	640.2
20	705.1	697.8	690.5	683.1	675.8	661.0	646.2
25	708.7	701.6	694.4	687.3	680.2	665.9	651.7
30	712.0	705.1	698.1	691.2	684.3	670.4	656.7
35	715.2	708.5	701.7	694.9	688.2	674.7	661.5
40	718.4	711.7	705.1	698.5	692.0	678.9	666.0

[a] $U(\rho)$ = 1.7 kg · m^{-3} (288.15 K ⩽ T ⩽ 363.15 K) and 2.7 kg · m^{-3} (373.15 K ⩽ T ⩽ 413.15 K) (expanded uncertainty with 0.95 level of confidence).
[b] $U(T)$ = 0.1 K (expanded uncertainty with 0.95 level of confidence).
[c] $U(p)$ = 0.05 MPa (expanded uncertainty with 0.95 level of confidence).
[d] $U(x) = 2 \cdot 10^{-4}$ (expanded uncertainty with 0.95 level of confidence).

Table 5
Experimental densities, ρ, for the binary mixture *n*-heptane (1) + ethanol (2) for different mole fractions, *x*, at pressure, *p*, and temperature, *T*.

p^c/MPa	ρ^a/kg · m^{-3}						
	n-Heptane (1) + ethanol (2)						
	T^b/K						
	293.15	303.15	313.15	323.15	333.15	353.15	373.15
x_1[d] = 0.2505							
0.1	737.7	728.9	719.9	710.6	701.1		
0.5	738.1	729.3	720.3	711.1	701.6	681.4	659.3
5	742.1	733.6	725.0	716.1	706.9	687.7	666.9
10	746.4	738.2	729.8	721.2	712.5	694.1	674.5
15	750.4	742.4	734.3	726.1	717.6	700.0	681.4
20	754.3	746.5	738.6	730.6	722.4	705.5	687.6
25	757.9	750.3	742.6	734.8	726.9	710.5	693.3
30	761.4	754.0	746.5	738.9	731.2	715.3	698.6
35	764.7	757.5	750.1	742.7	735.2	719.7	703.6
40	768.0	760.8	753.7	746.4	739.1	724.0	708.4
x_1 = 0.5004							
0.1	711.0	702.1	693.0	683.6	674.0		
0.5	711.3	702.5	693.4	684.1	674.5	654.2	632.2
5	715.4	706.9	698.2	689.2	680.0	660.8	640.2
10	719.8	711.5	703.1	694.5	685.7	667.5	648.1
15	723.8	715.8	707.7	699.4	691.0	673.6	655.3
20	727.7	719.9	712.0	704.0	695.9	679.1	661.7
25	731.3	723.8	716.1	708.4	700.5	684.3	667.5
30	734.8	727.4	720.0	712.4	704.8	689.1	672.9
35	738.1	730.9	723.7	716.3	708.9	693.7	678.0
40	741.4	734.3	727.2	720.1	712.8	698.1	682.9
x_1 = 0.7501							
0.1	694.3	685.4	676.4	667.2	657.7		
0.5	694.6	685.8	676.9	667.6	658.2	638.4	617.2
5	698.7	690.3	681.6	672.8	663.8	645.1	625.3
10	703.0	694.9	686.6	678.1	669.6	651.8	633.3
15	707.1	699.2	691.2	683.0	674.9	657.9	640.4
20	710.9	703.3	695.5	687.7	679.8	663.5	646.8
25	714.6	707.1	699.6	692.0	684.3	668.6	652.6
30	718.0	710.8	703.4	696.0	688.6	673.4	658.0
35	721.4	714.3	707.1	699.9	692.7	678.0	663.0
40	724.6	717.6	710.6	703.6	696.6	682.3	667.9

[a] $U(\rho)$ = 1.7 kg · m^{-3} (288.15 K ⩽ T ⩽ 363.15 K) and 2.7 kg · m^{-3} (373.15 K ⩽ T ⩽ 413.15 K) (expanded uncertainty with 0.95 level of confidence).
[b] $U(T)$ = 0.1 K (expanded uncertainty with 0.95 level of confidence).
[c] $U(p)$ = 0.05 MPa (expanded uncertainty with 0.95 level of confidence).
[d] $U(x) = 2 \cdot 10^{-4}$ (expanded uncertainty with 0.95 level of confidence).

$$\rho(T,p) = \frac{\rho^{\mathrm{ref}}(T)}{1 - C(T)\ln\left(\frac{B(T)+p}{B(T)+p^{\mathrm{ref}}}\right)} \tag{1}$$

$$\rho^{\mathrm{ref}}(T) = \sum_{i=0}^{2} a_i T^i \tag{2}$$

$$B(T) = \sum_{i=0}^{2} b_i T^i \tag{3}$$

$$C(T) = \sum_{i=0}^{2} c_i T^i \tag{4}$$

where ρ^{ref} is density at reference pressure p^{ref} that is chosen to be 0.1 MPa. In our case the parameter C is treated as temperature independent.

The quality of density data modeling was verified using the absolute average percentage deviation, *AAD*, and the standard deviation, σ [16]:

$$AAD = \frac{100}{N}\sum_{i=1}^{N}\left|\frac{\rho_i^{\mathrm{exp}} - \rho_i^{\mathrm{cal}}}{\rho_i^{\mathrm{exp}}}\right| \tag{5}$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}\left(\rho_i^{\mathrm{exp}} - \rho_i^{\mathrm{cal}}\right)^2}{N-m}} \tag{6}$$

where, ρ^{exp} is the experimentally determined density, ρ^{cal} refers to the density values obtained using the Eq. (1), N represents the number of experimental data and m is the number of the optimized parameters of the modified Tammann–Tait equation. Tables 7 and 8 collect adjusted parameters and the corresponding deviations obtained using the modified Tammann–Tait equation for the pure compounds and binary mixtures, respectively. Very good agreement between experimental and calculated densities of the pure compounds (Figs. S1 and S2) is confirmed with low *AAD* values, 0.04% for *n*-heptane, 0.03% for *n*-octane and 0.08% for ethanol. Additionally, experimental densities of the binary mixtures agree very well with the calculated densities with *AAD*s not higher than 0.02%.

3.3. Derived thermodynamic properties calculation

From density data valuable derived properties can be calculated. Here, excess molar volume, V^{E}, isothermal compressibility, κ_T, and isobaric thermal expansion coefficient, α_p, were determined.

Table 6
Experimental densities, ρ, for the binary mixture *n*-octane (1) + ethanol (2) for different mole fractions, *x*, at pressure, *p*, and temperature, *T*.

p^c/MPa	ρ^a/kg · m^{-3}						
	n-Octane (1) + ethanol(2)						
	T^b/K						
	293.15	303.15	313.15	323.15	333.15	353.15	373.15
x_1[d] = 0.2492							
0.1	743.9	735.3	726.5	717.4	708.2		
0.5	744.3	735.7	726.9	717.9	708.6	689.0	667.6
5	748.2	739.8	731.3	722.6	713.7	695.0	674.8
10	752.3	744.2	736.0	727.6	719.1	701.1	682.0
15	756.2	748.3	740.4	732.2	724.0	706.8	688.6
20	759.9	752.2	744.5	736.6	728.6	712.0	694.6
25	763.4	755.9	748.4	740.7	733.0	716.9	700.1
30	766.8	759.5	752.1	744.6	737.1	721.5	705.2
35	770.0	762.9	755.7	748.4	741.0	725.8	710.0
40	773.2	766.2	759.1	752.0	744.8	730.0	714.6
x_1 = 0.4999							
0.1	722.3	713.7	705.0	696.0	686.8		
0.5	722.6	714.1	705.4	696.4	687.3	667.9	647.2
5	726.5	718.3	709.8	701.2	692.4	674.0	654.4
10	730.6	722.6	714.5	706.1	697.7	680.2	661.7
15	734.4	726.7	718.8	710.8	702.7	685.8	668.3
20	738.1	730.6	722.9	715.1	707.3	691.1	674.3
25	741.6	734.2	726.8	719.2	711.6	695.9	679.8
30	744.9	737.7	730.5	723.1	715.7	700.5	684.9
35	748.1	741.1	734.0	726.8	719.6	704.8	689.7
40	751.2	744.3	737.4	730.4	723.4	709.0	694.3
x_1 = 0.7503							
0.1	709.6	701.2	692.6	683.8	674.8		
0.5	709.9	701.5	693.0	684.2	675.3	656.6	637.0
5	713.7	705.6	697.4	688.9	680.4	662.6	644.1
10	717.7	709.9	702.0	693.8	685.6	668.7	651.2
15	721.5	713.9	706.3	698.4	690.5	674.3	657.7
20	725.1	717.8	710.3	702.7	695.1	679.5	663.6
25	728.6	721.4	714.1	706.8	699.3	684.3	669.0
30	731.9	724.9	717.8	710.6	703.4	688.7	674.0
35	735.0	728.2	721.2	714.2	707.2	693.0	678.7
40	738.1	731.4	724.6	717.8	710.9	697.1	683.2

[a] $U(\rho)$ = 1.7 kg · m^{-3} (288.15 K ⩽ T ⩽ 363.15 K) and 2.7 kg · m^{-3} (373.15 K ⩽ T ⩽ 413.15 K) (expanded uncertainty with 0.95 level of confidence).
[b] $U(T)$ = 0.1 K (expanded uncertainty with 0.95 level of confidence).
[c] $U(p)$ = 0.05 MPa (expanded uncertainty with 0.95 level of confidence).
[d] $U(x) = 2 \cdot 10^{-4}$ (expanded uncertainty with 0.95 level of confidence).

$$V^E = \sum_{i=1}^{n} x_i M_i \left[\left(\frac{1}{\rho}\right) - \left(\frac{1}{\rho_i}\right) \right] \tag{7}$$

In Eq. (7) x_i denotes the mole fraction of a component *i* in a mixture; M_i its molecular weight and ρ and ρ_i are the measured densities of a mixture and a pure component *i*, respectively.

Isothermal compressibility describes the pressure effect on density at constant temperature as follows

$$\kappa_T = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial p}\right)_T \tag{8}$$

Incorporation of the Eq. (1) in Eq. (8) leads to [4]

$$\kappa_T = \frac{C(T)}{(B(T)+p)\left(1 - C(T)\ln\frac{B(T)+p}{B(T)+p^{ref}}\right)} \tag{9}$$

Isobaric thermal expansion coefficient expresses the temperature influence on a fluid density at constant pressure and can be calculated by using the following equation

$$\alpha_p = \frac{1}{\rho}\left(\frac{\partial p}{\partial T}\right)_\rho \left(\frac{\partial p}{\partial \rho}\right)_T^{-1} = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_p \tag{10}$$

The following expression can be derived combining Eqs. (1) and (10) [4]

$$\alpha_p = \frac{-\rho^{ref\prime}(T)}{\rho^{ref}(T)} - \frac{C(T)\frac{B'(T)(p^{ref}-p)}{(B(T)+p)(B(T)+p^{ref})} + C'(T) ln\frac{B(T)+p}{B(T)+p^{ref}}}{\left(1 - C(T) ln\frac{B(T)+p}{B(T)+p^{ref}}\right)} \tag{11}$$

where $\rho^{ref\prime}(T)$, $B'(T)$ and $C'(T)$ are derivatives with respect to T of parameters ρ^{ref}, $B(T)$ and $C(T)$, respectively

$$\rho^{ref\prime} = \sum_{i=0}^{2} i a_i T^{i-1} \tag{12}$$

$$B' = \sum_{i=0}^{2} i b_i T^{i-1} \tag{13}$$

$$C' = \sum_{i=0}^{2} i c_i T^{i-1} \tag{14}$$

As confirmed by Abdulagatov and Azizov [11], system *n*-heptane + *n*-octane exhibits small deviations from ideal behavior. With increasing temperature negative excess volume also increases, while with pressure raise a mixture approaches ideal behavior [11]. This was expected due to similar shape and structure of *n*-heptane and *n*-octane molecules and its low polarity.

Fig. 4 depicts changes of excess molar volume of equimolar *n*-heptane + ethanol and *n*-octane + ethanol mixtures vs. pressure for different isotherms, while in Fig. S3, changes of V^E vs. composition of *n*-alkane at selected pressures and temperatures are presented for the same mixtures. Both mixtures containing alcohol show positive deviation from ideal behavior. As it has been discussed before [12,35] deviation from ideal behavior might be a consequence of many different factors that occur upon mixing, e.g. the association of dissimilar compounds through hydrogen bonding, dipole–dipole interactions, dispersion forces, interstitial accommodation of the molecules due to a larger differences in molar volumes, steric hindrance of the molecules, etc. It might be assumed that in the analyzed mixtures positive contributions are predominantly attributed to hydrogen bond rupture and stretching of self-associated molecules of alcohol [35,36]. Namely, ethanol molecules form aggregates through hydrogen bonds. Infrared spectra in conjunction with heat of mixing data for ethanol + *n*-heptane solutions at temperatures between 293.15 and 323.15 K reveal three possible hydrogen-bond types for ethanol in *n*-heptane solution, i.e. monomeric alcohol units, double-bonded dimmers and linear polymer chains [37]. The addition of *n*-heptane or *n*-octane to pure ethanol provokes the weakening or breaking of hydrogen bonds in ethanol self-associates and an increase in the free volume [10]. Moreover, the temperature raise increases the thermal energy of molecules and furthermore weakens the associations between like molecules. Both figures confirm that the temperature raise significantly influences a raise in positive excess volume values, noticeably at (353.15 and 373.15) K. Contrarily, the pressure increase causes a mixture compression, e.g. excess volume decreases, which might be due to an increase of the association constant of the alcohol [9] and due to the shortening of the distance between compounds and decrease in the free volume [10]. However, the influence of pressure is more pronounced at lower pressures, up to around 20 MPa. At higher pressures excess molar volumes approaches constant value for a specific temperature (see Fig. 4).

The isothermal compressibility, κ_T, and the isobaric thermal expansion coefficient, α_p, of pure compounds at various temperatures and pressures are presented in Tables S1 and S2, respectively, and in Figs. S4–S6. Figs. 5–7 depict changes of isothermal compressibility, (Table S3), and isobaric thermal expansion coefficient (Table S4) with pressure at selected temperatures (293.15 and 323.15 K) for the analyzed binary mixtures. As Figs. S4–S6 and

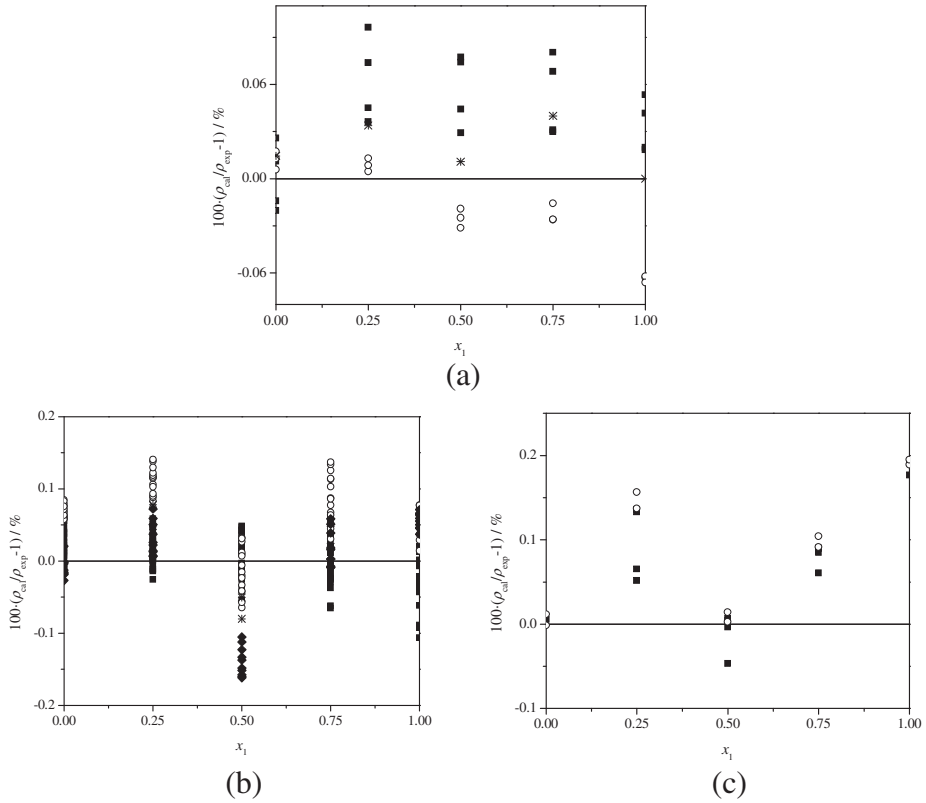


Fig. 2. Comparison between measured densities and those previously reported against mole fraction (x_1) for (a) *n*-heptane (1) + *n*-octane: (■) Abdulagatov and Azizov [11] at (303.15–333.15) K, (○) Matos et al. [22] at (293.15–313.15) K and (✳) Al Gherwi et al. [33] at 313.15 K, all at atmospheric pressure; (b) *n*-heptane (1) + ethanol: (■) Watson et al. [10] at (293.15–333.15) K up to 40 MPa; (○) Dzida and Marczak [9] at (293.15–313.15) K up to 40 MPa, (◆) Papaioannou et al. [7] at 298.15 K up to 33.8 MPa and (✳) Orge et al. [34] at 303.15 K and 313.15 K at atmospheric pressure, and (c) *n*-octane (1) + ethanol: (■) Estrada-Baltazar et al. [26] at (293.15–323.15) K and (○) Orge et al. [34] at 303.15 K and 313.15 K, all at atmospheric pressure.

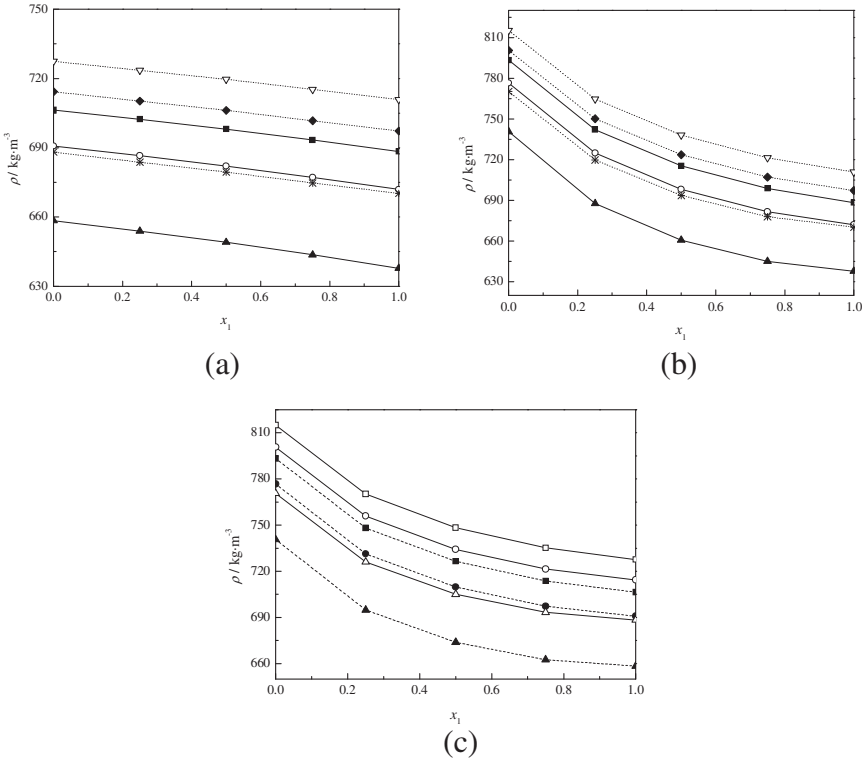


Fig. 3. Experimental densities versus mole fraction for the (a) *n*-heptane (1) + *n*-octane (2), (b) *n*-heptane (1) + ethanol (2) and (c) *n*-octane (1) + ethanol systems at 5 MPa (full symbols) and 35 MPa (empty symbols) for selected isotherms: (■, □) 293.15 K, (●, ○) 313.15 K, and (▲, △) 353.15 K. Lines are guides for eyes.

Table 7
Adjusted parameters of Eqs. (1)–(4), standard deviation, σ, and average percentage deviation, *AAD*, for the pure compounds.

	n-Heptane	*n*-Octane	Ethanol
a_0/kg · m^{-3}	871.86	875.98	952.55
a_1/kg · m^{-3} · K^{-1}	−0.44006	−0.39170	−0.26716
a_2/kg · m^{-3} · K^{-2}	$-6.8295 \cdot 10^{-4}$	$-6.8294 \cdot 10^{-4}$	$-9.8674 \cdot 10^{-4}$
b_0/MPa	364.98	376.90	521.31
b_1/MPa · K^{-1}	−1.4465	−1.4388	−2.0143
b_2/MPa · K^{-2}	$1.4864 \cdot 10^{-3}$	$1.4222 \cdot 10^{-3}$	$2.1155 \cdot 10^{-3}$
c_0	0.090856	0.091360	0.11571
σ/kg · m^{-3}	0.464	0.343	1.143
AAD/%	0.044	0.032	0.075

5–7 show, isothermal compressibility and isobaric thermal expansion coefficient of the pure compounds decrease in the following order: *n*-heptane > *n*-octane > ethanol within the investigated p, T range. Both κ_T and α_p increase with temperature increase and decrease as pressure increases. However, the pressure influence on κ_T for the *n*-heptane and *n*-octane is rather higher than for ethanol. It is assumed that structure of ethanol is more rigid due to hydrogen bonded OH groups [9]. Also, the temperature influence on isothermal compressibility reduces as pressure increases. α_p isotherms for pure *n*-heptane and *n*-octane do not intersect at investigated pressure range, but show tendency to intersect at pressures slightly above 60 MPa (see Figs. S4 and S5). At the

Table 8
Adjusted parameters of Eqs. (1)–(4), standard deviation, σ, and average percentage deviation, *AAD*, for the binary mixtures.

x_1	*n*-Heptane (1) + *n*-octane			*n*-Heptane (1) + ethanol			*n*-Octane(1) + ethanol		
	0.2500	0.5002	0.7502	0.2505	0.5004	0.7501	0.2492	0.4999	0.7503
a_0/kg · $^{-3}$	886.86	881.92	874.49	858.601	835.52	837.22	868.53	853.178	859.71
a_1/kg · $^{-3}$ K^{-1}	−0.47668	−0.46306	−0.43618	0.031042	0.015346	−0.11263	−0.012068	−0.057854	−0.19781
a_2/kg · m^{-3} · K^{-2}	$-5.6905 \cdot 10^{-4}$	$-6.0833 \cdot 10^{-4}$	$-6.7036 \cdot 10^{-4}$	$-1.5131 \cdot 10^{-3}$	$-1.5024 \cdot 10^{-3}$	$-1.2798 \cdot 10^{-3}$	$-1.4095 \cdot 10^{-3}$	$-1.3262 \cdot 10^{-3}$	$-1.0726 \cdot 10^{-3}$
b_0/MPa	330.45	342.71	343.22	366.94	362.02	353.74	361.14	361.06	336.83
b_1/MPa K^{-1}	−1.2758	−1.3482	−1.359	−1.4233	−1.4373	−1.4117	−1.3741	−1.4022	−1.2815
b_2/MPa · K^{-2}	$1.2751 \cdot 10^{-3}$	$1.3768 \cdot 10^{-3}$	$1.3930 \cdot 10^{-3}$	$1.4250 \cdot 10^{-3}$	$1.4653 \cdot 10^{-3}$	$1.4457 \cdot 10^{-3}$	$1.3480 \cdot 10^{-3}$	$1.4003 \cdot 10^{-3}$	$1.2440 \cdot 10^{-3}$
c	0.082865	0.084243	0.085089	0.089459	0.0875	0.086668	0.088070	0.085702	0.084095
σ/kg · m^{-3}	0.202	0.206	0.210	0.304	0.278	0.266	0.281	0.237	0.214
AAD/%	0.012	0.012	0.014	0.022	0.020	0.019	0.020	0.017	0.014

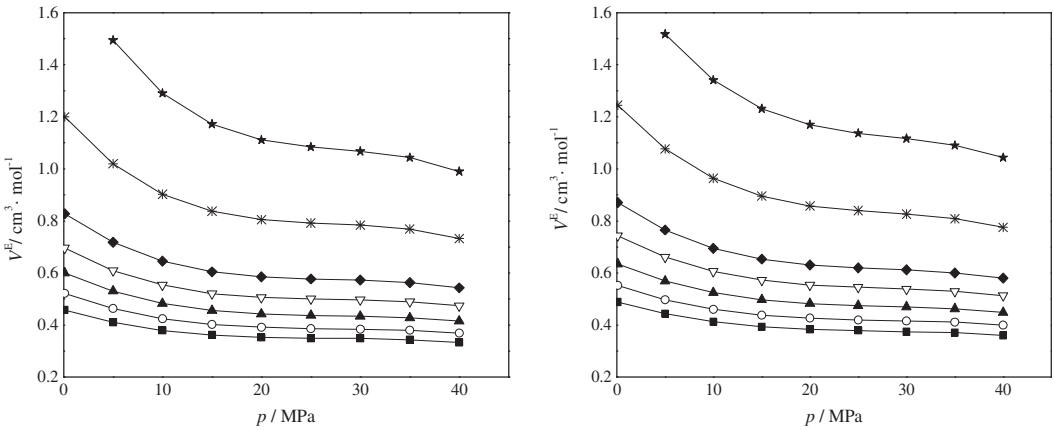


Fig. 4. Excess molar volume (V^E) vs. pressure (p) for the equimolar binary mixtures: (a) *n*-heptane (1) + ethanol (2) and (b) *n*-octane (1) + ethanol (2) at different isotherms: (■) 293.15 K, (○) 303.15 K, (▲) 313.15 K, (▽) 323.15 K, (◆) 333.15 K, (✳) 353.15 K, (★) 373.15 K

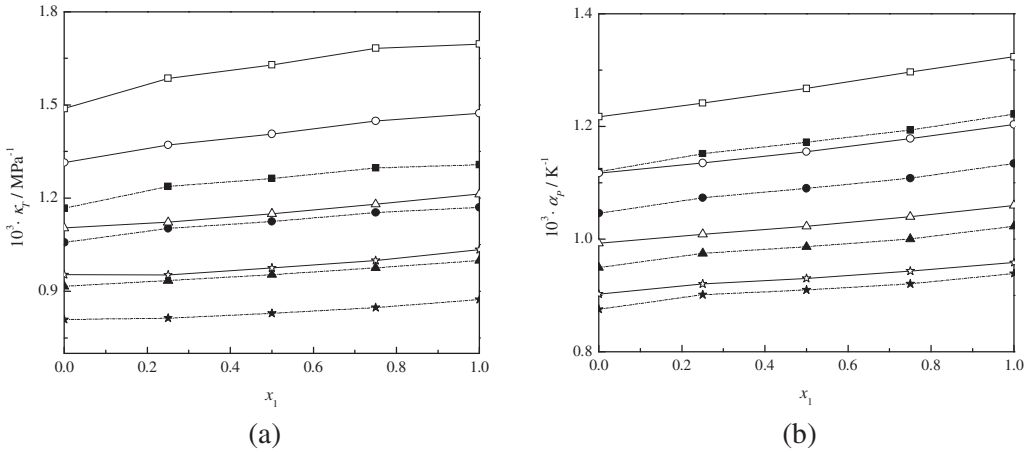


Fig. 5. Isothermal compressibility, κ_T, (a) and isobaric thermal expansion coefficient, α_p, (b) vs. mole fraction of *n*-heptane (x_1) for *n*-heptane (1) + *n*-octane (2) system at at 293.15 K (full symbols) and 323.15 K (empty symbols) for selected isobars: (■, □) 0.1 MPa, (●, ○) 10 MPa, (▲, △) 25 MPa and (★, ☆) 40 MPa. Lines are guides for eyes.

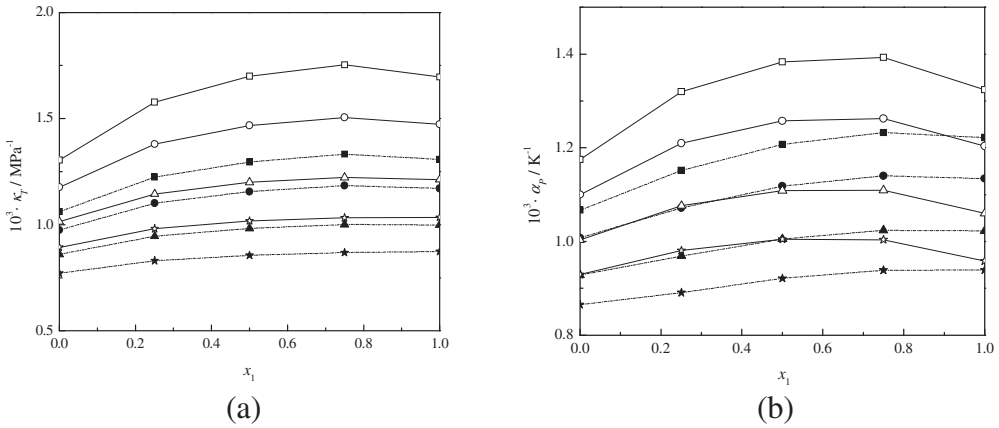


Fig. 6. Isothermal compressibility, κ_T, (a) and isobaric thermal expansion coefficient, α_p, (b) vs. mole fraction of *n*-heptane (x_1) for *n*-heptane (1) + ethanol (2) system at 293.15 K (full symbols) and 323.15 K (empty symbols) for selected isobars: (■, □) 0.1 MPa, (●, ○) 10 MPa, (▲, △) 25 MPa and (★, ☆) 40 MPa. Lines are guides for eyes.

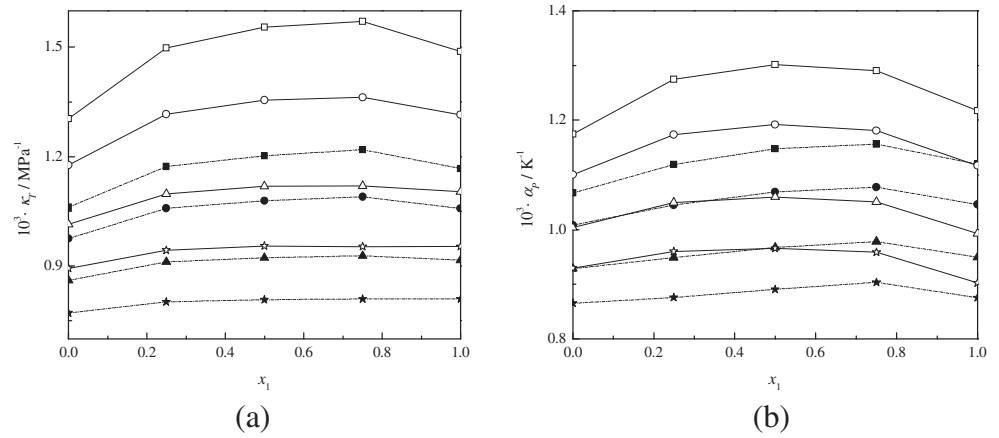


Fig. 7. Isothermal compressibility, κ_T, (a) and isobaric thermal expansion coefficient, α_p, (b) vs. mole fraction of *n*-octane (x_1) for *n*-octane (1) + ethanol (2) system at 293.15 K (full symbols) and 323.15 K (empty symbols) for selected isobars: (■, □) 0.1 MPa, (●, ○) 10 MPa, (▲, △) 25 MPa and (★, ☆) 40 MPa. Lines are guides for eyes.

pressures corresponding to the "crossing point" isobaric thermal expansivity becomes temperature independent, while at pressures higher than at crossing point the temperature influence on α_p becomes opposite [5]. For the pure ethanol it can be presumed that the crossing point could appear at very high pressures (Fig. S6) which is in accordance with similar behavior confirmed for other alcohols [38]. It is evident that κ_T and α_p vs. x curves have different shapes for *n*-heptane + *n*-octane system comparing to *n*-alkane + alcohol systems. As the amount of *n*-heptane increases in the mixture with *n*-octane both κ_T and α_p increase almost linearly (see Fig. 5). It is also evident, that κ_T and α_p vs. x curves for *n*-heptane + ethanol and *n*-octane + ethanol at 323.15 K show non-monotonical behavior (Figs. 6 and 7). *n*-Octane is slightly more expansive than ethanol at lower pressures and temperatures, but at higher temperatures and pressures becomes less expansive than ethanol. This might be due to the fact that α_p behavior of *n*-alkanes is mainly dominated by dispersive interactions [39] which are less sensitive to temperature change than hydrogen bonging. In addition, the structure of ethanol is stabilized by hydrogen bonds at lower temperatures. Increase in temperature influences hydrogen bond destruction and ethanol as well as the solution becomes more elastic. This effect diminishes as pressure rises due to the stronger hydrogen bonds, i.e. the pressure prevents breaking of hydrogen bonds [9]. This behavior is in accordance with pressure and temperature influence on positive excess volume data already explained.

4. Conclusion

Densities, excess molar volumes, isothermal compressibilities and isobaric thermal expansion coefficients for the pure *n*-heptane, *n*-octane and ethanol and for binary mixtures of *n*-heptane + *n*-octane, *n*-heptane + ethanol and *n*-octane + ethanol were determined. Experiments were conducted at pressures up to 60 MPa for the pure compounds and up to 40 MPa for the binary mixtures. Experimental densities were fitted by the modified Tammann–Tait equation. Very good agreement between experimental and calculated densities was confirmed with low standard and average percentage deviations. Also, comparison of the experimental and literature density values demonstrated a good agreement. Small excess molar volumes in the binary *n*-heptane + *n*-octane systems confirmed almost ideal behavior. On the other hand, volume expansion and higher isobaric thermal expansion coefficient or isothermal compressibilities occur for *n*-heptane + ethanol and *n*-octane + ethanol mixtures than in the pure compounds. These effects are more pronounced at elevated temperatures. It is assumed that positive contributions are predominantly attributed

to hydrogen bond rupture and stretching of self-associated molecules of ethanol, what is enhanced by temperature increase. Contrarily, the pressure raise causes a mixture compression since it prevents breaking of hydrogen bonds and increases the association constant of an alcohol.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jct.2016.04.014.

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